

Transistor-Level CMOS Design and Optimization of a 4-Bit Array Multiplier using Tanner EDA

Interim Report

1. You have to implement an array multiplier that performs the multiplication of two four-bit numbers, $A = A_3A_2A_1A_0$ and $B = B_3B_2B_1B_0$ as part of this project. The product of A and B is represented by the eight-bit number, $P = P_7P_6P_5P_4P_3P_2P_1P_0$. The circuit diagram of the array multiplier is shown in Fig. 1. You have to follow the steps given below to complete the project.

(a) Implement reference two-input AND, OR, and XOR gates as the basic building blocks of the multiplier. Plot the VTC and find out the propagation delays of the gates assuming that all inputs transition simultaneously

Ans: **For AND GATE**

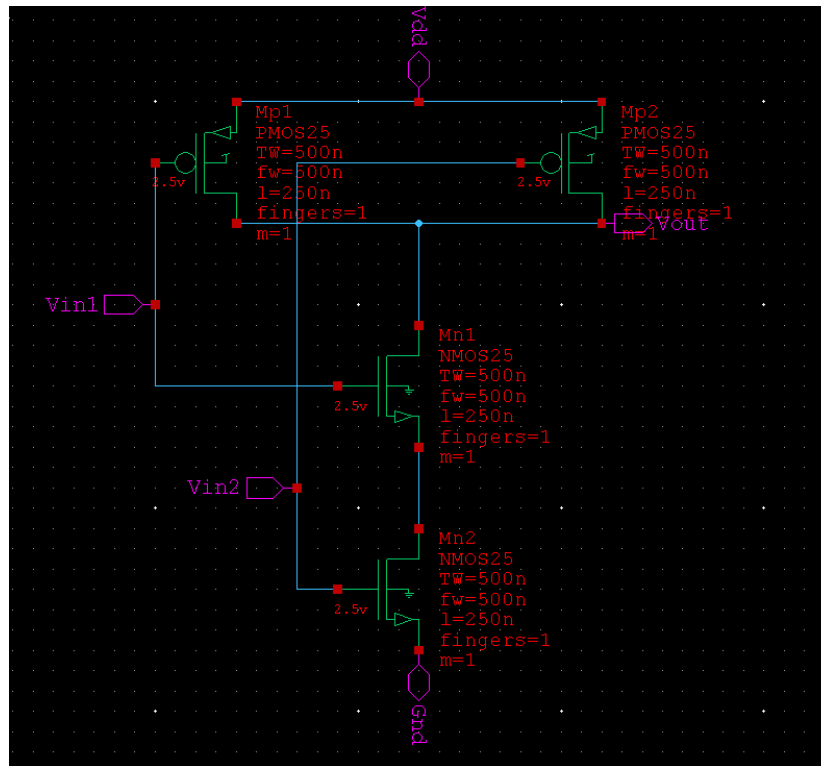


Figure-1 NAND gate

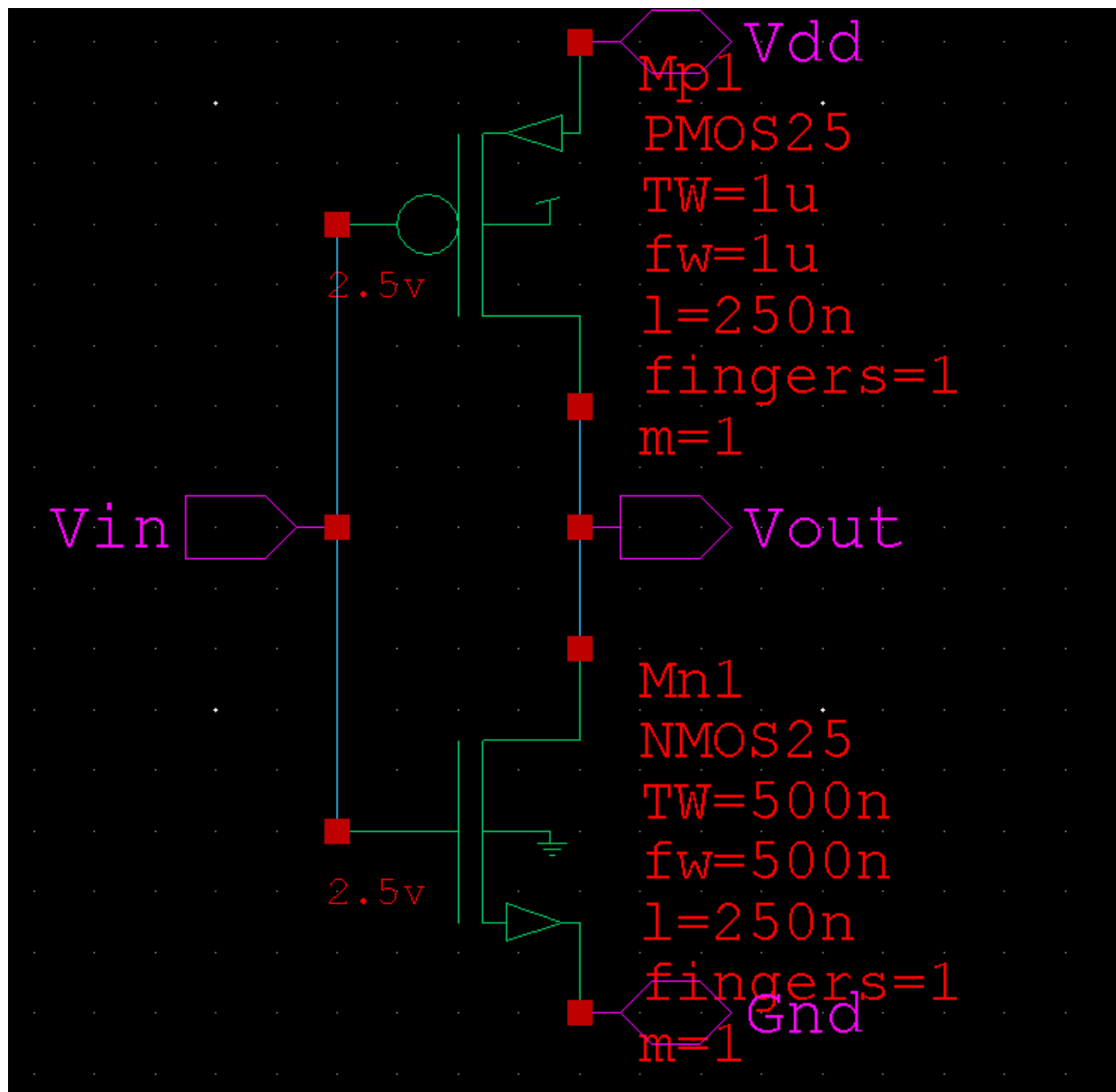


Figure-2 Inverter

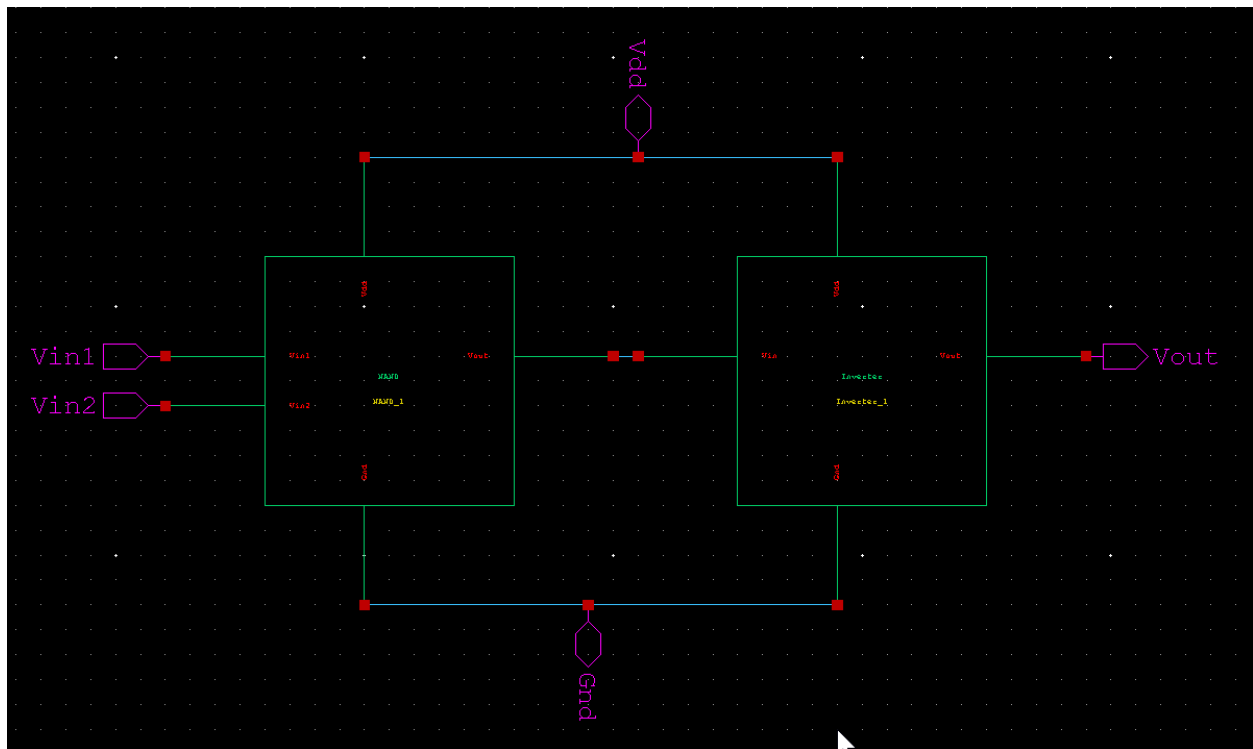
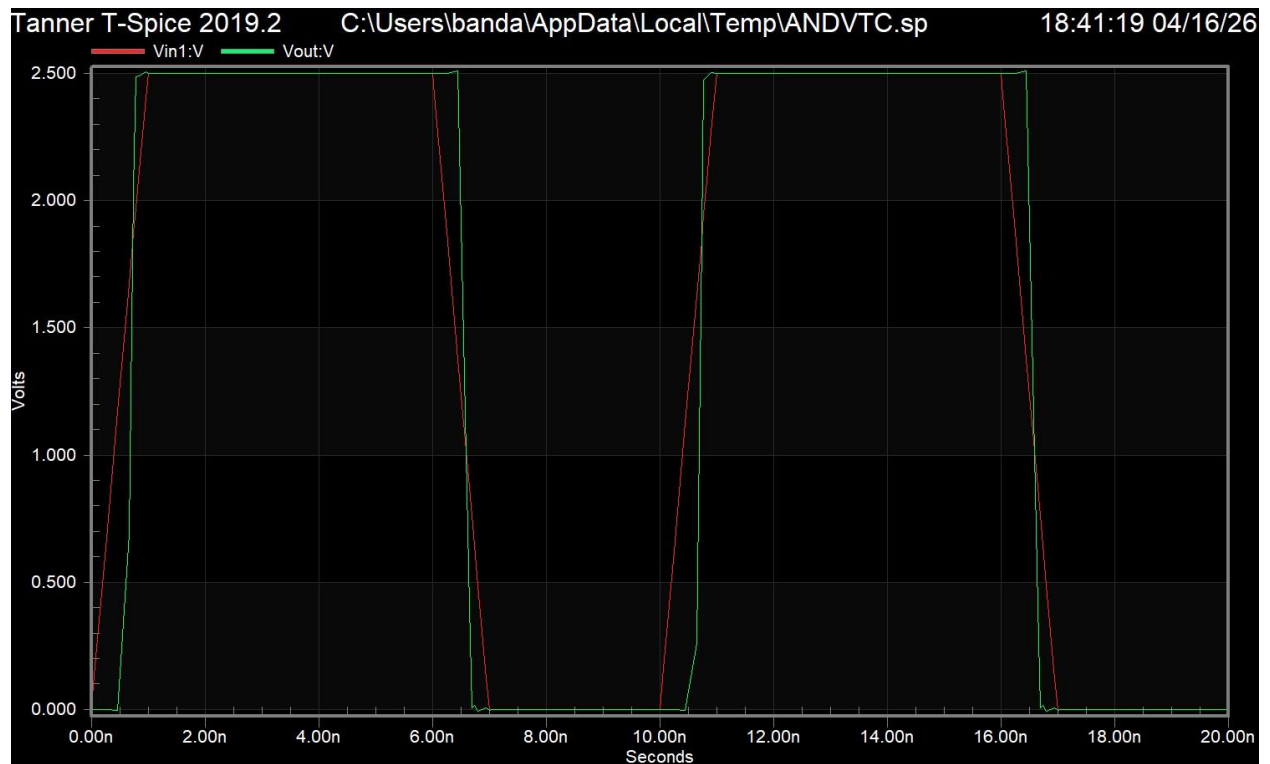


Figure-3 AND Gate

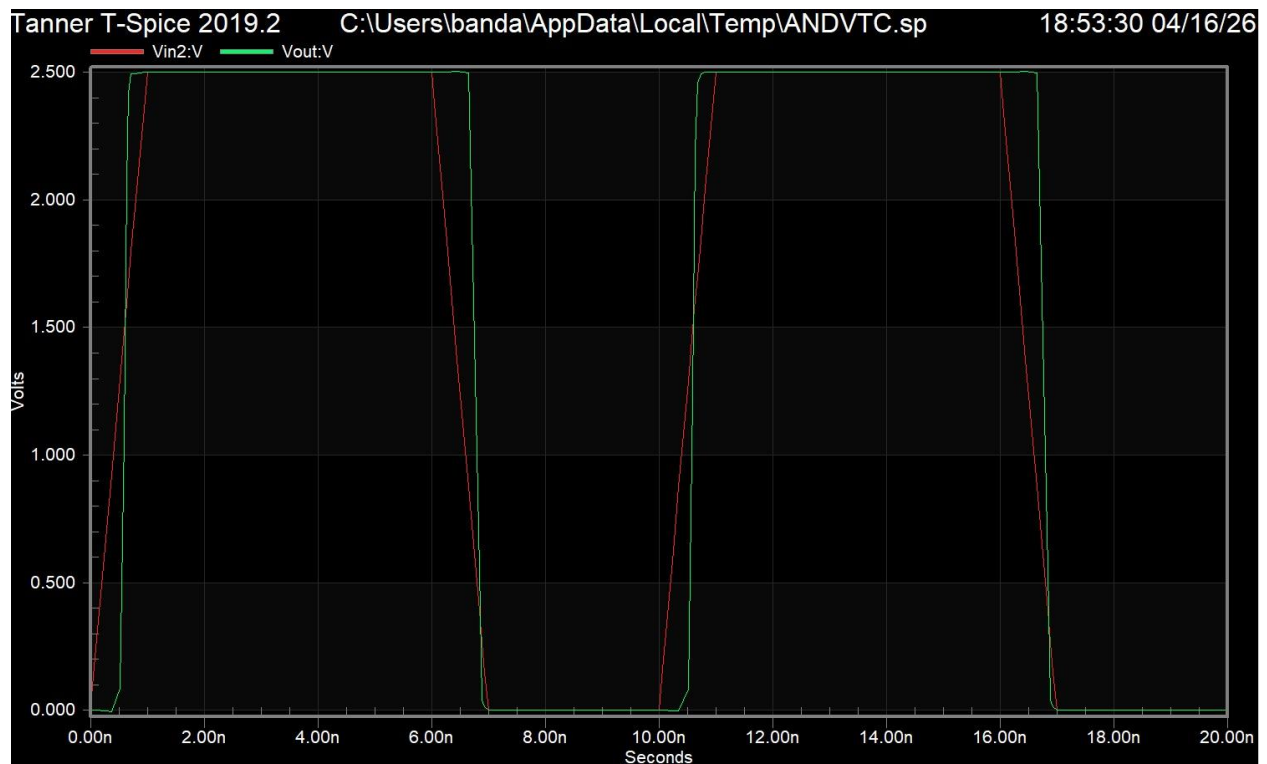


VTC of AND Gate For a=b=0-1

Propagation delays of the gates

$T_{pHL}=61\text{ps}$

$t_{pLH}=178\text{ps}$



VTC of AND Gate For $a=1$ $b=0-1$

Propagation delays of the gates

$t_{pHL}=279\text{ps}$

$t_{pLH}=70\text{ps}$



VTC of AND Gate For a=0 b=0-1

Propagation delays of the gates

tpHL=0

tpLH=0

For OR Gate

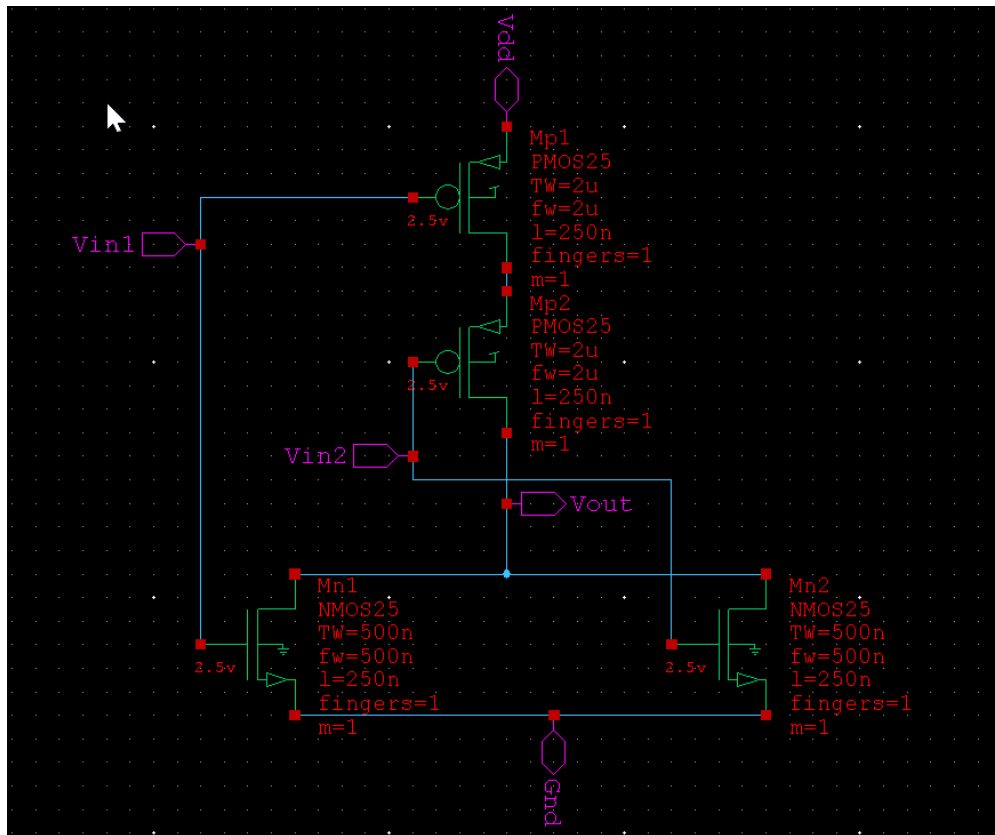


Figure NOR Gate

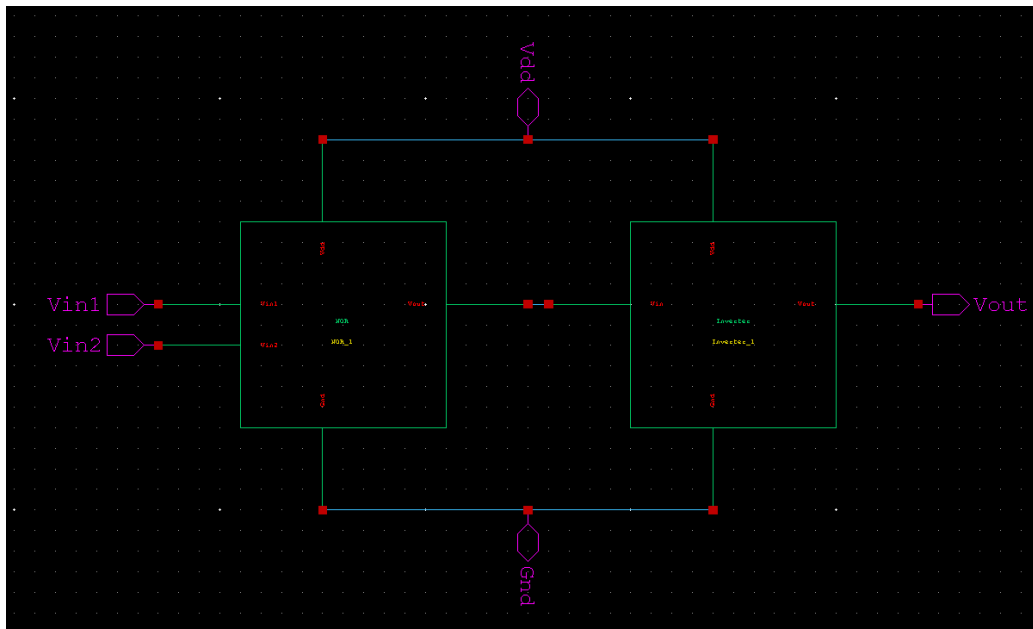
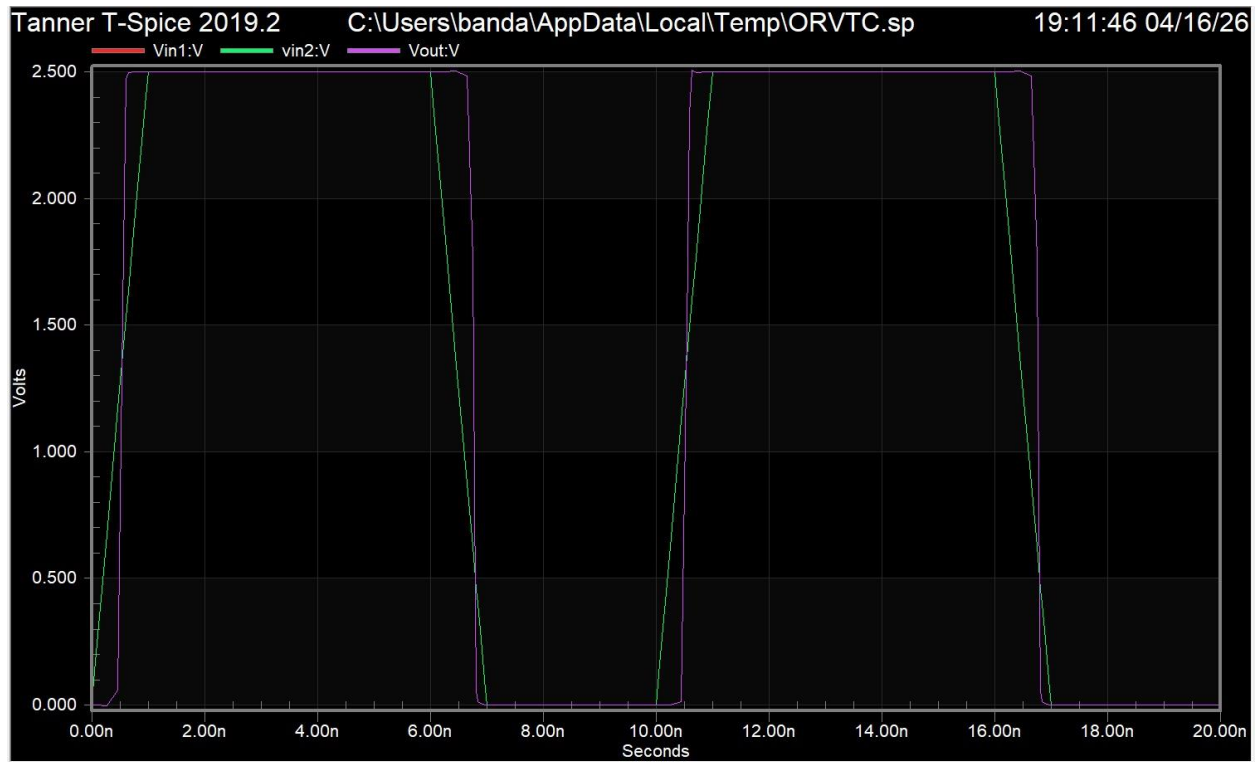


Figure OR Gate



VTC of OR Gate For a=b=0-1

Propagation delays of the gates

tpHL=295ps

tpLH=56ps

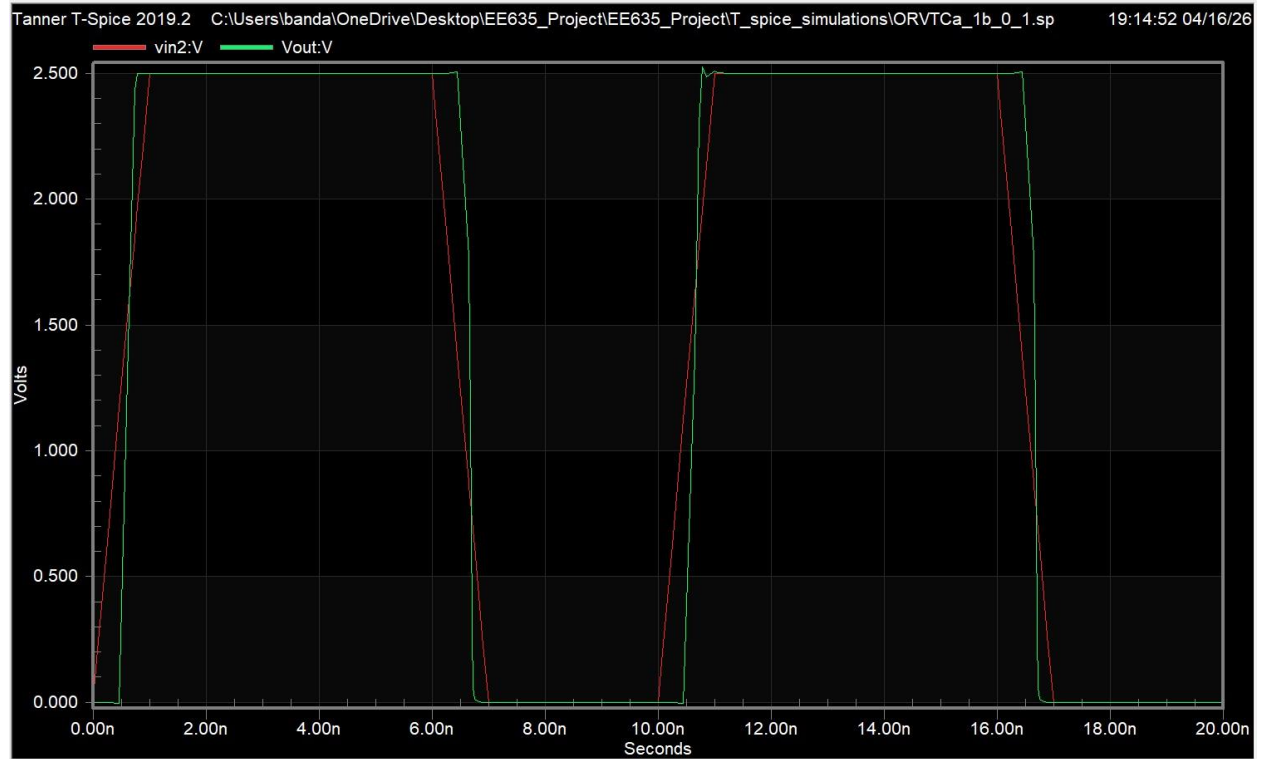


VTC of OR Gate For a=0 b=0-1

Propagation delays of the gates

tpHL=0

tpLH=0



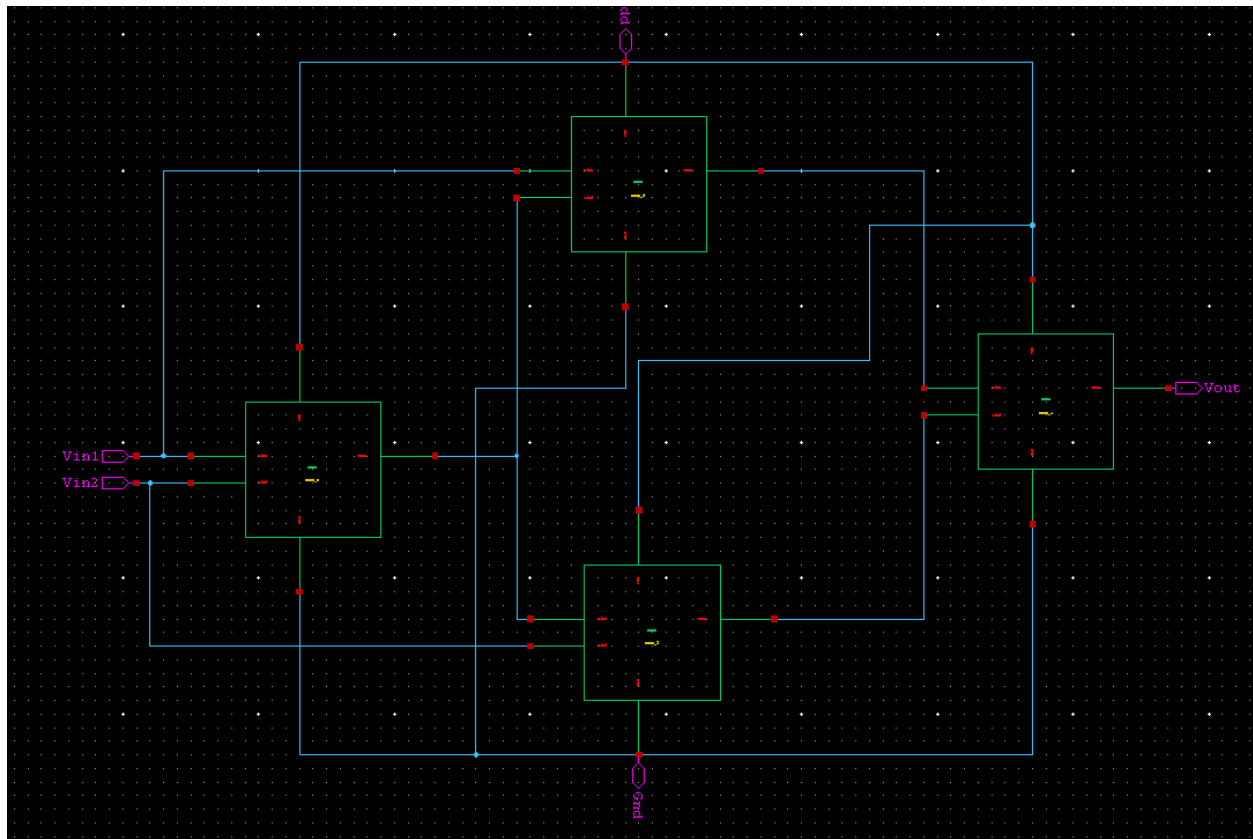
VTC of OR Gate For a=0 b=0-1

Propagation delays of the gates

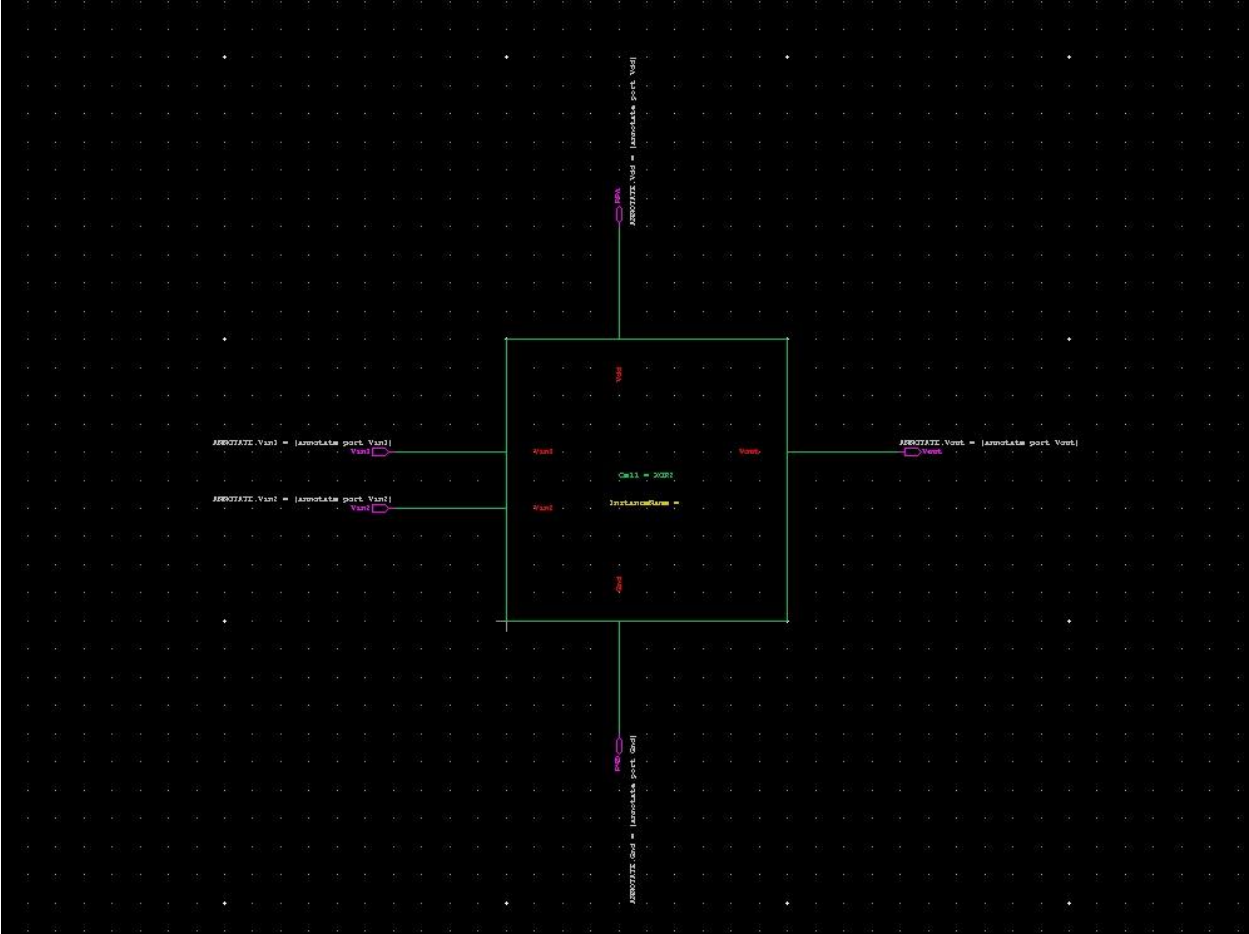
tpHL=0.18ns

tpLH=0.15ns

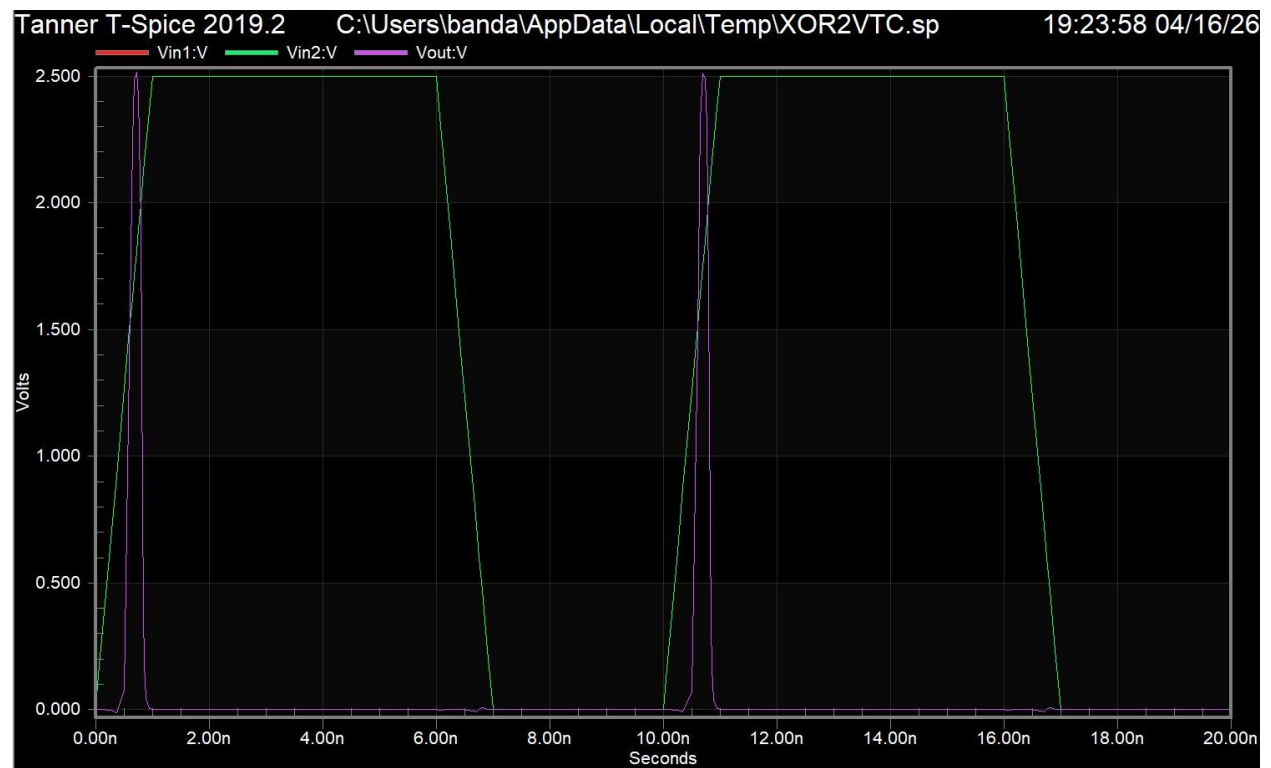
For XOR Gate



For XOR Gate using NAND Gate's



Schematic Of XOR Gate

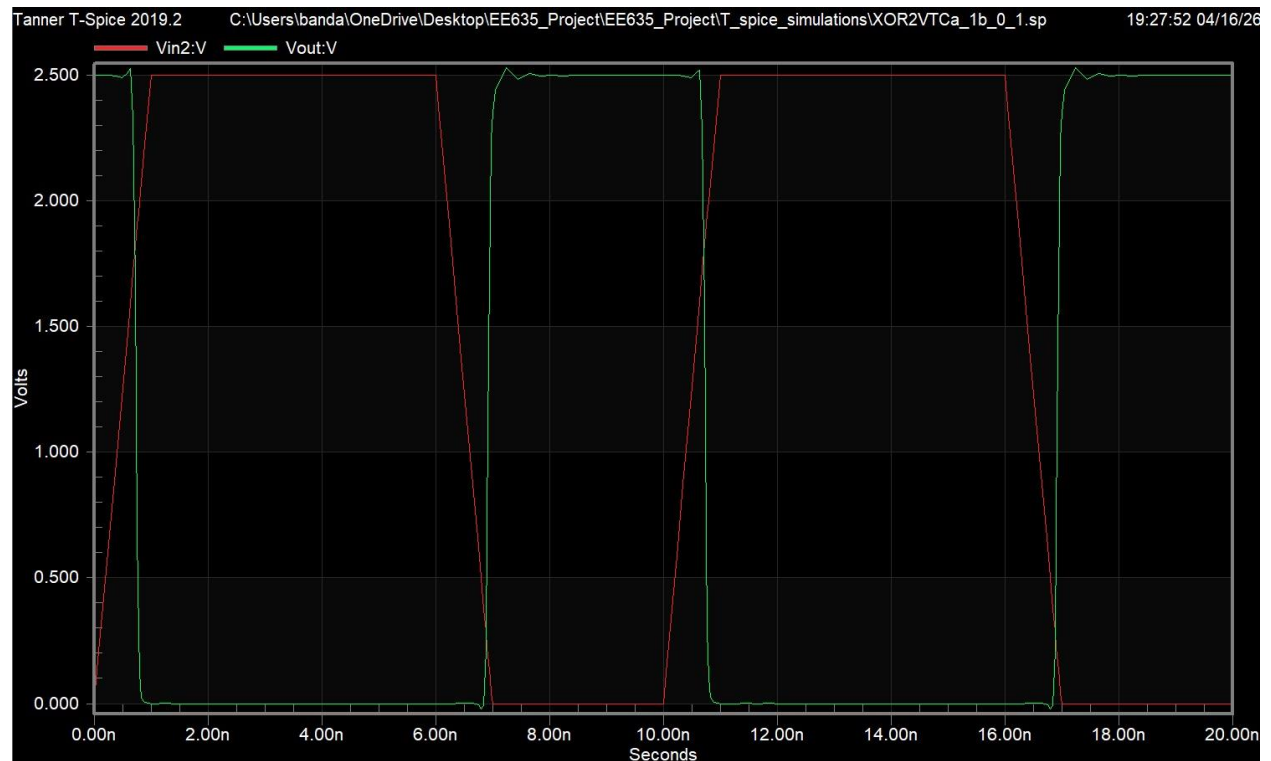


VTC of XOR Gate a=b=0-1

Propagation delays of the gates

tpHL=0

tpLH=0

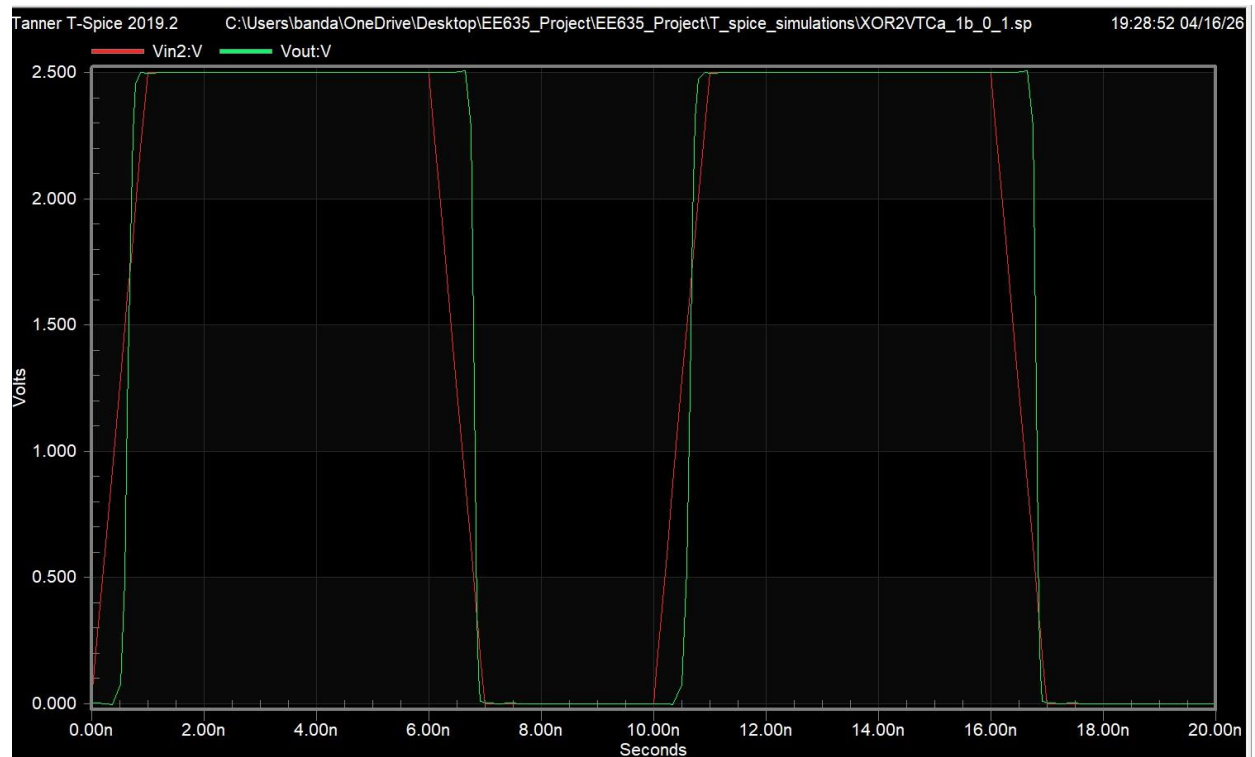


VTC of XOR Gate $a=1$ $b=0-1$

Propagation delays of the gates

$t_{pHL}=308\text{ps}$

$t_{pLH}=370\text{ps}$



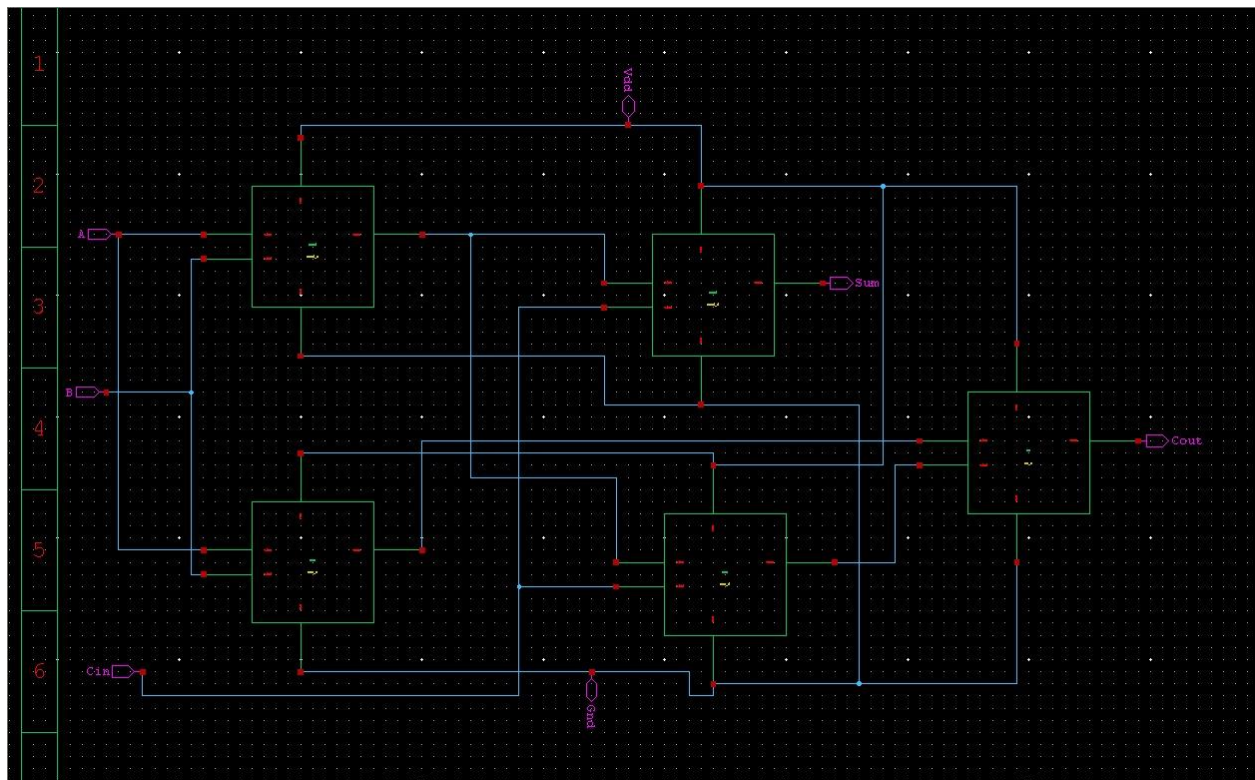
VTC of XOR Gate $a=0$ $b=0-1$

Propagation delays of the gates

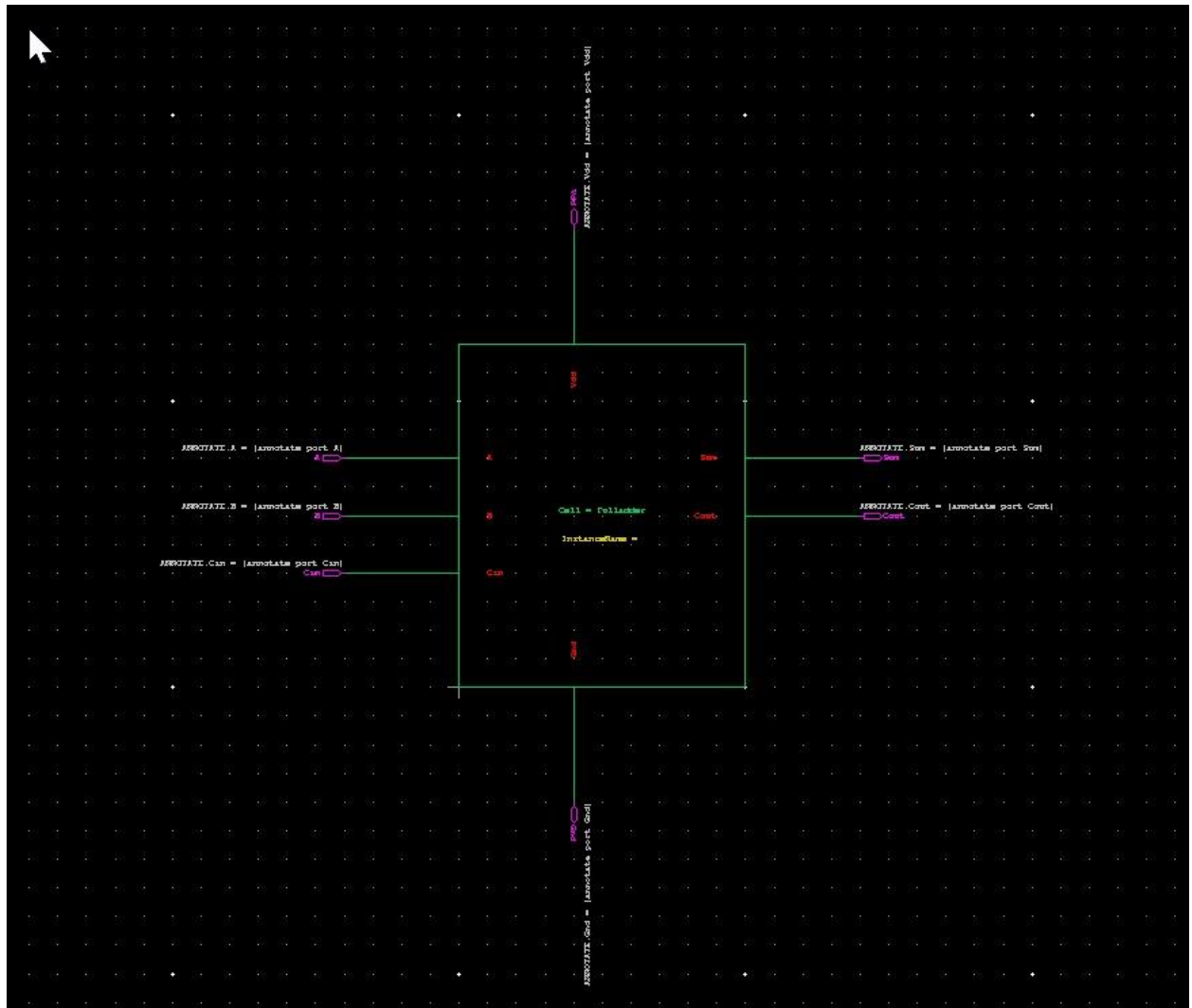
$t_{pHL}=0.2ns$

$t_{pLH}=0.2ns$

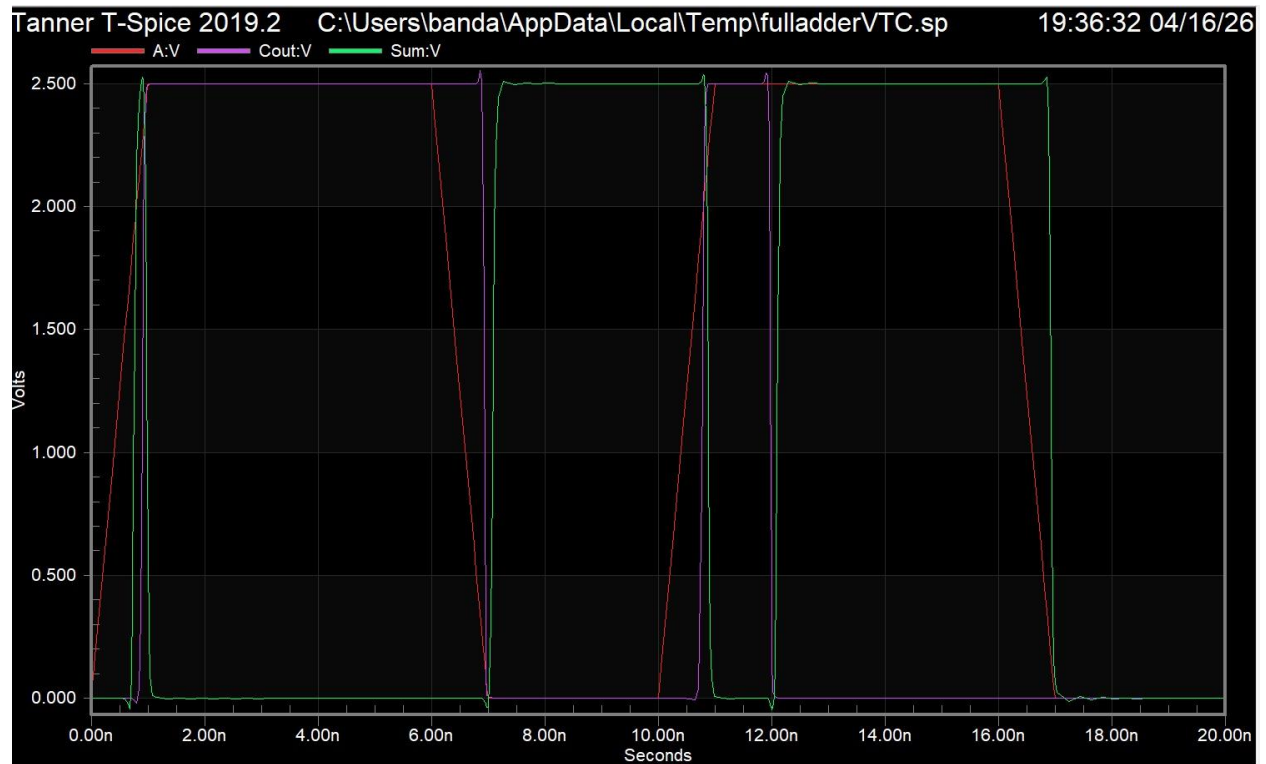
(b) Instantiating the AND, OR, and XOR gates, implement a full adder. Plot the VTC and find out the propagation delays of the full adder assuming that all inputs transition simultaneously.



Full adder schematic



Full Adder Symbol

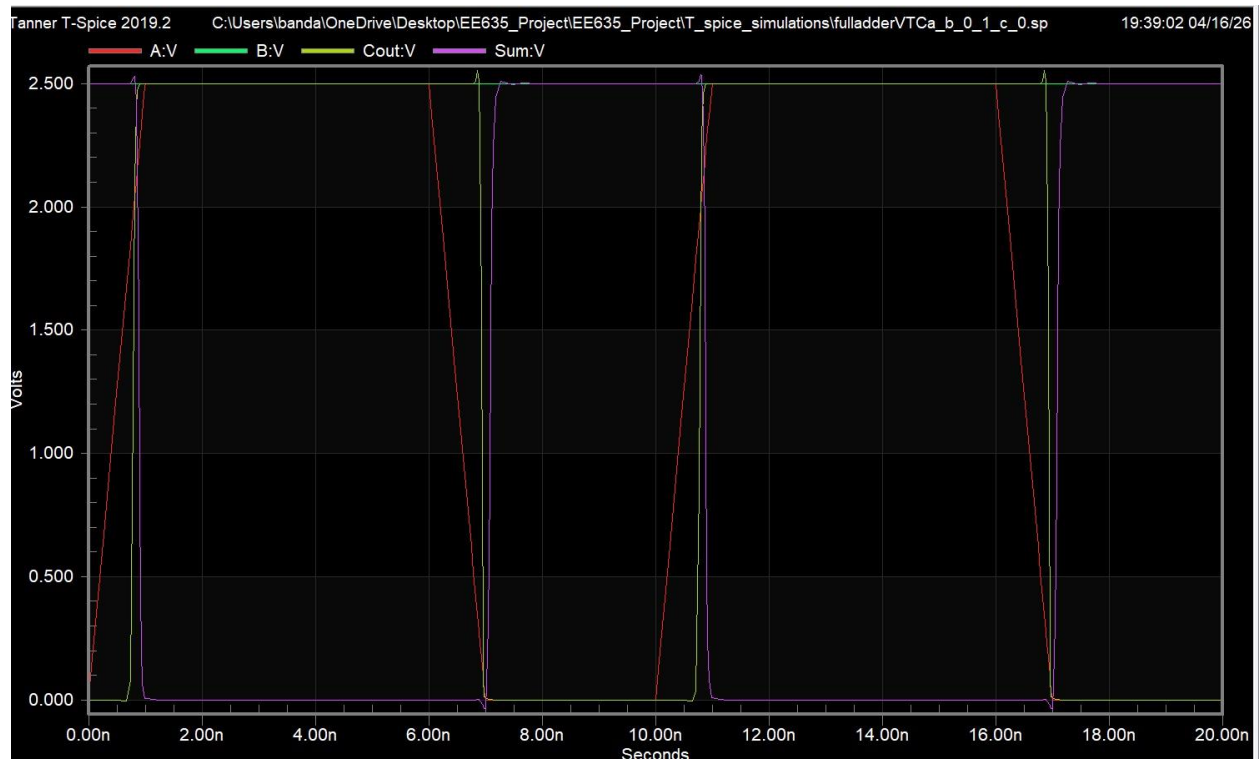


VTC Of Full Adder A=B=0->1 Cin=0

Propagation delays of the gates

tpHL=0.35ns

tpLH=0.3ns



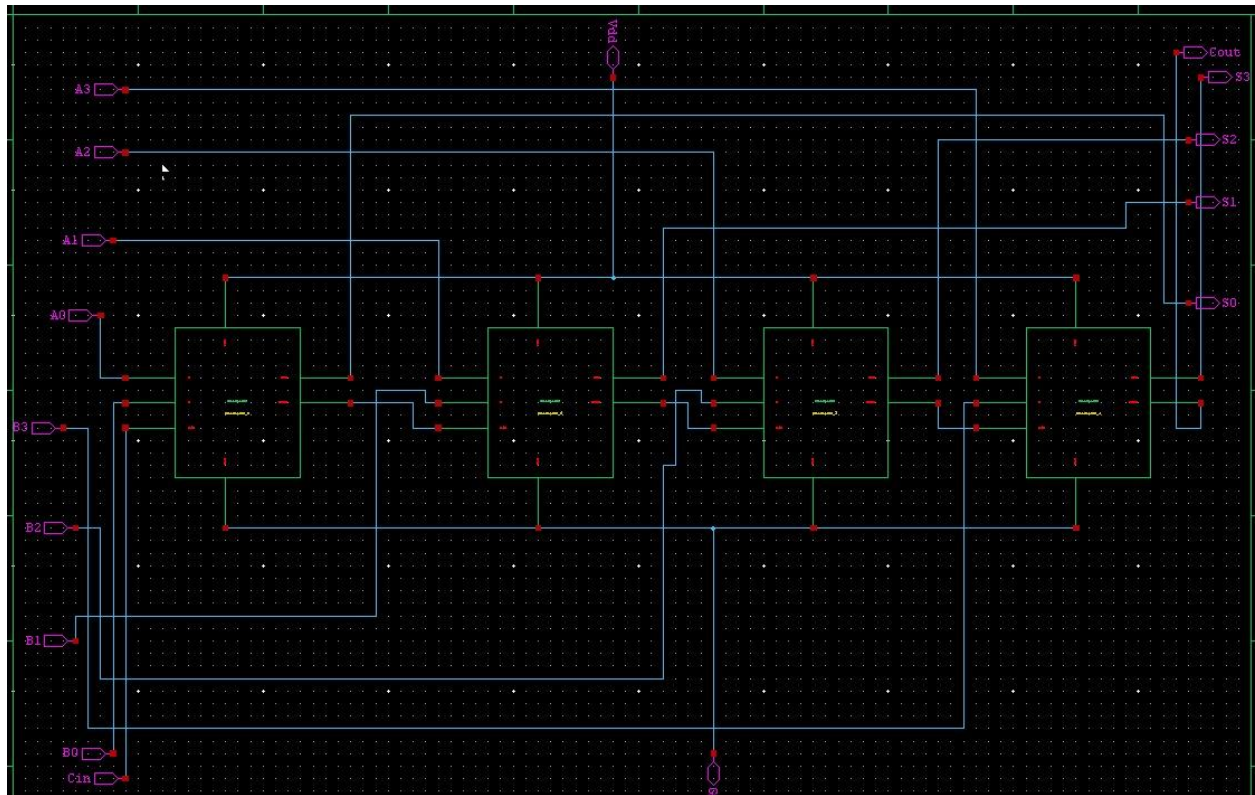
VTC of Full adder A=0-1 B=1 Cin=0

Propagation delays of the gates

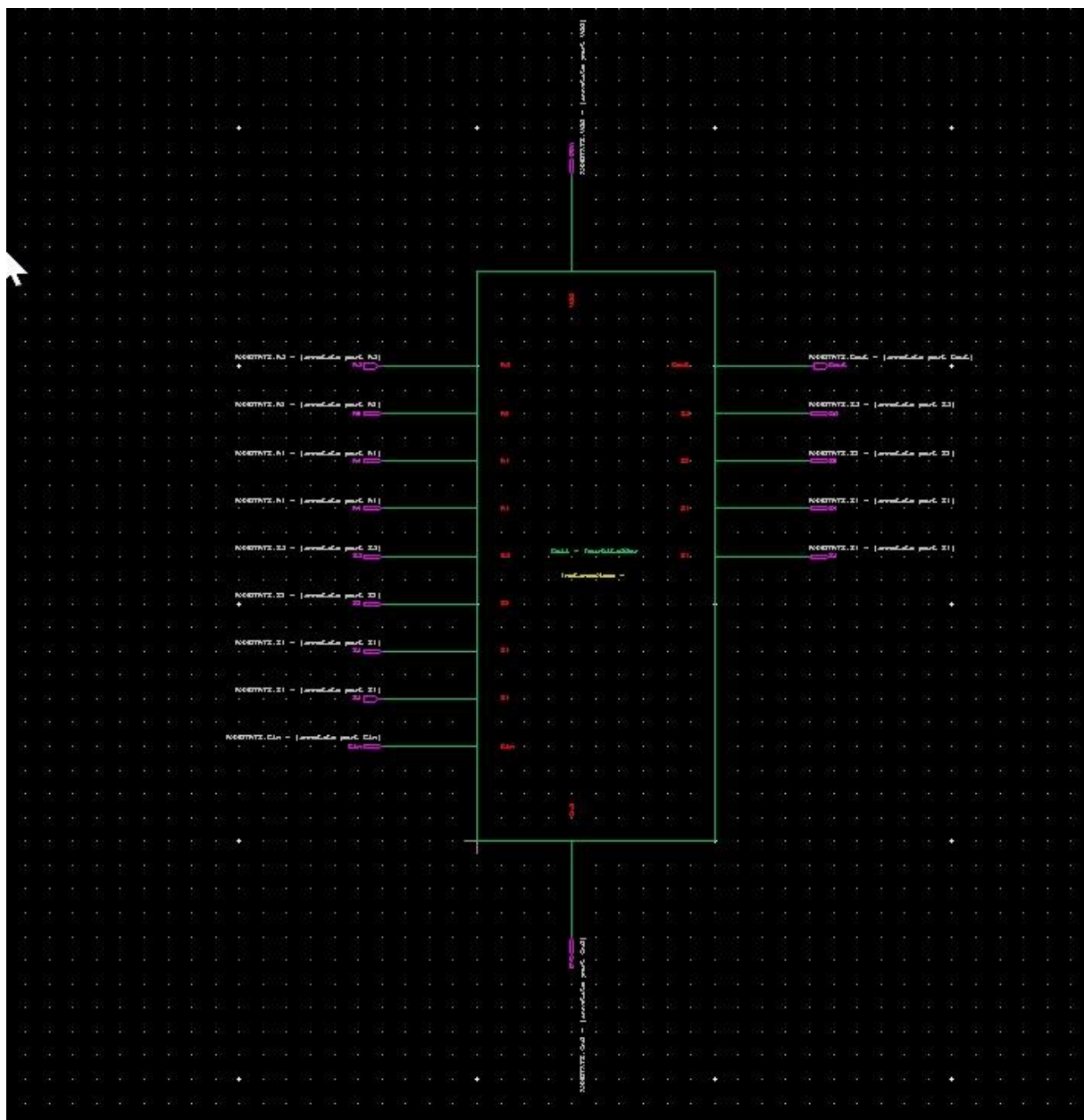
$t_{pHL}=0.3\text{ns}$

$t_{pLH}=0.25\text{ns}$

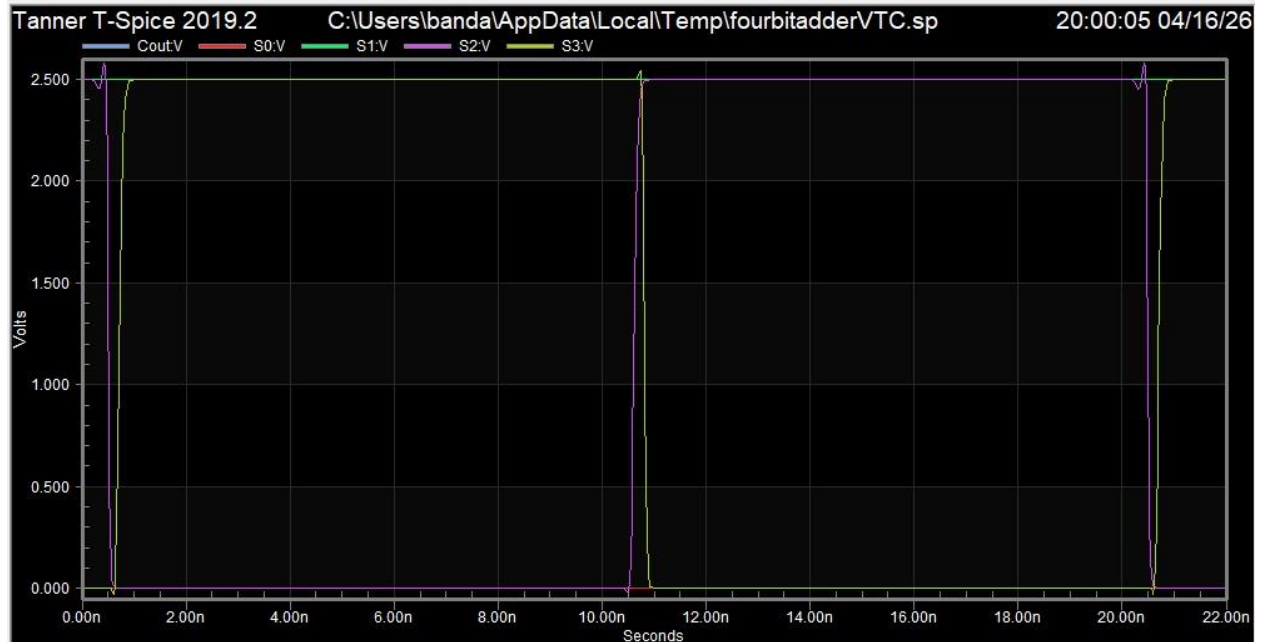
(c) Build a four-bit adder by instantiating the above full adder four times. Demonstrate the functionality through waveforms for 2-3 different input combinations. Estimate the critical path delay through simulations



Schematic of Four bit adder



Symbol of four bit addder



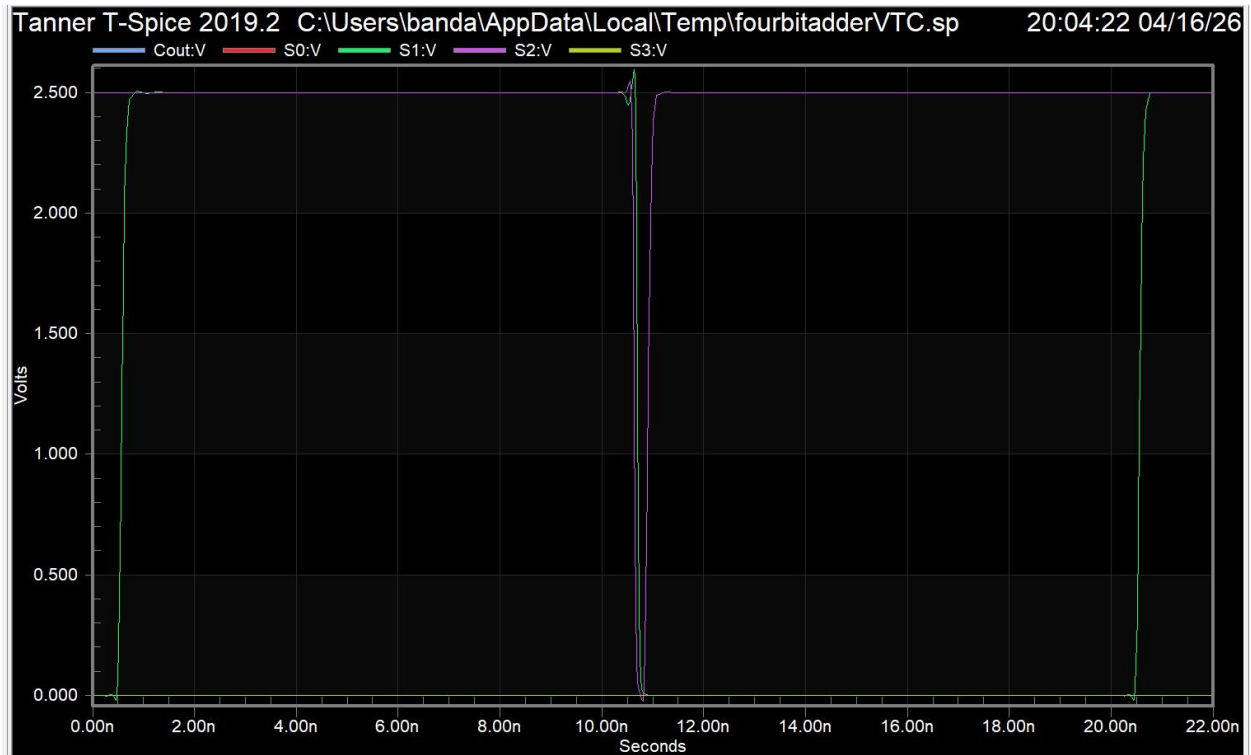
Case-1 A=0011 B=0111->0011 Cin=0

S=1010 ->0110

Cout=0

In above the critical path is B2 --> FA2 --> FA3 --> S3

Critical path delay is $t_{cd} = 668.6439\text{ps}$



Case-1 A=0011->0001 B=0011 Cin=0

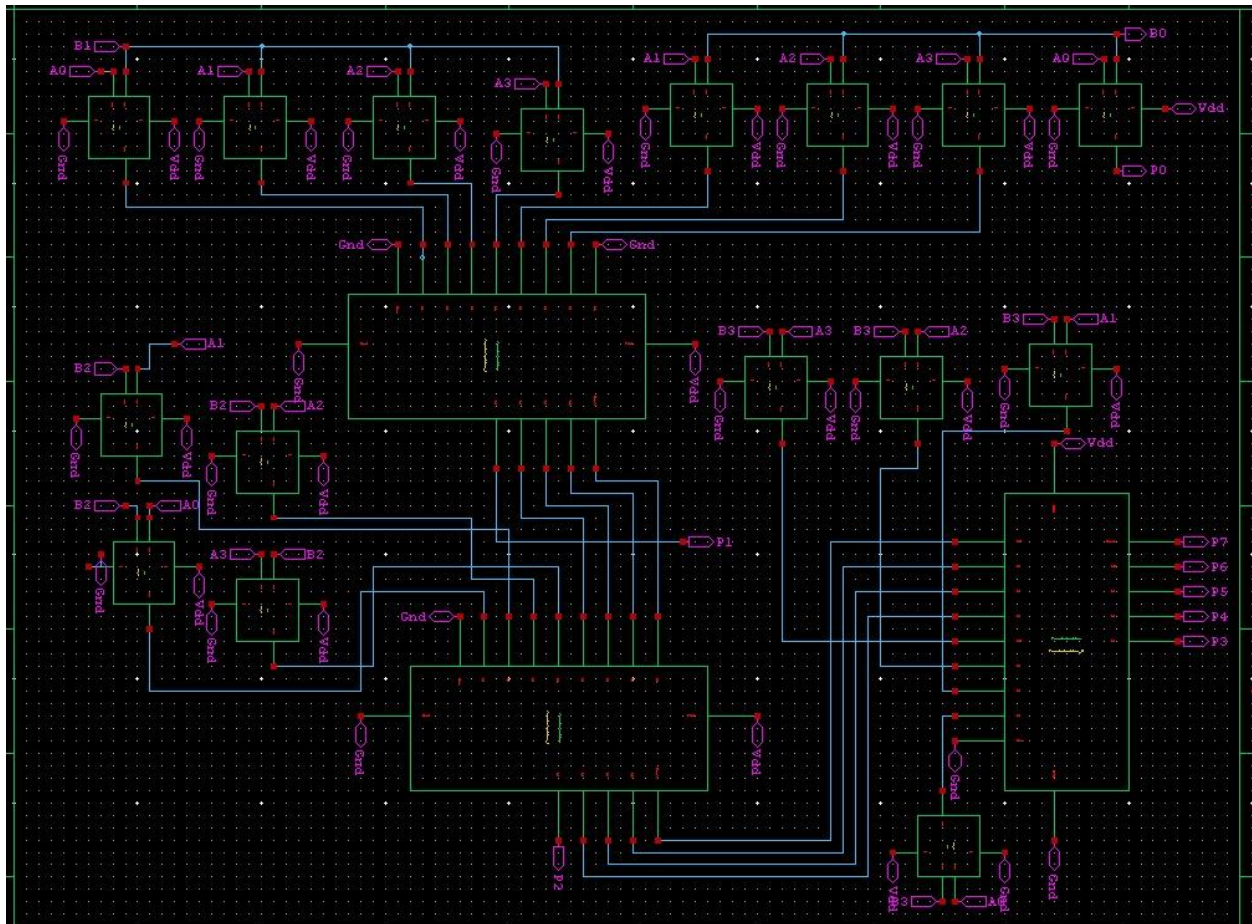
S=0110->0100

Cout=0

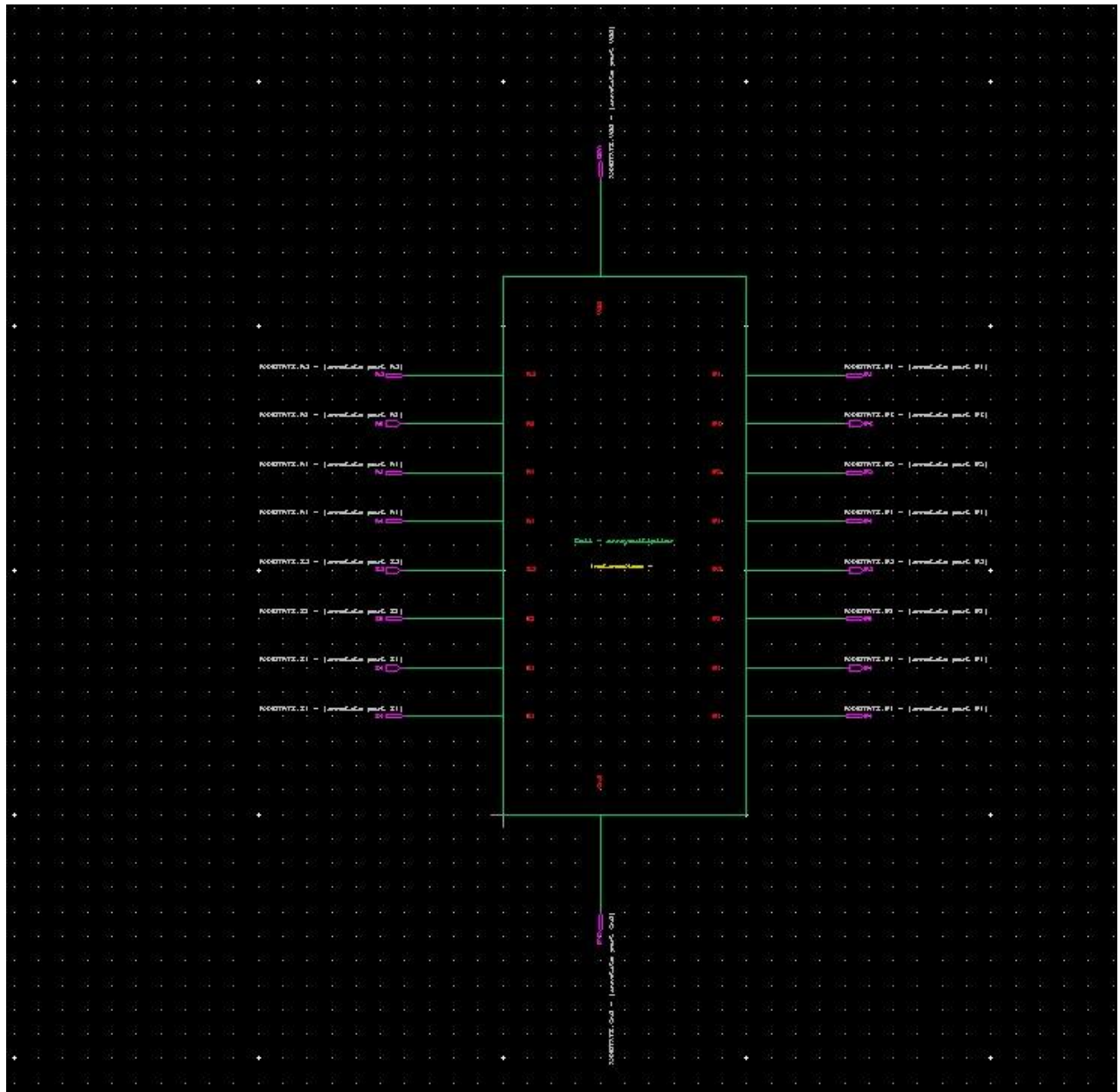
In above the critical path is A1 --> FA1 --> FA2 --> S2

Critical path delay is $t_{cd} = 524.5630\text{ps}$

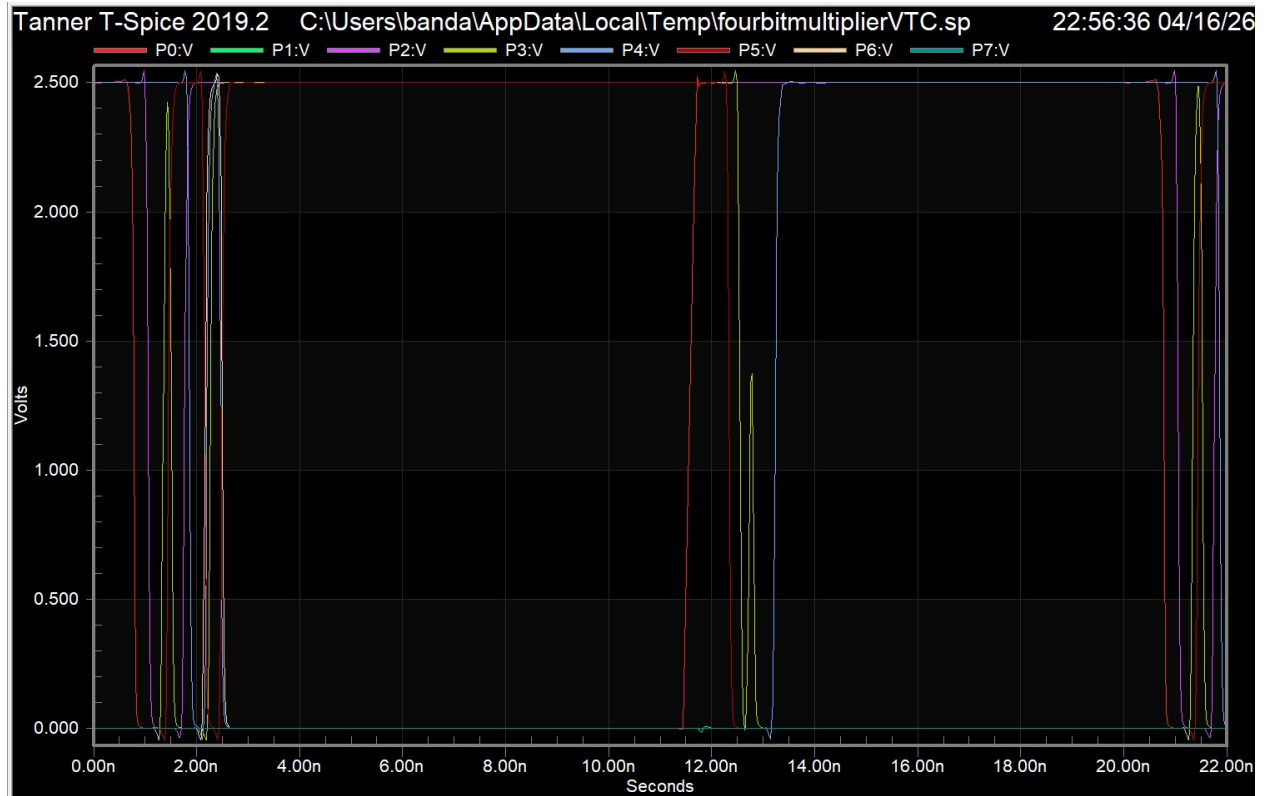
(d) Implement an array multiplier by instantiating the four-bit adder and AND gates as shown in Fig. 1. Demonstrate the functionality through waveforms for 2-3 different input combinations. Estimate the critical path delay through simulations. Calculate the power consumption and the transistor area for this implementation.



Schematic of Four bit array multiplier



Symbol of Four bit array multiplier



Input A=0111 -->1011

Input B=0011-->0100

Output P= 0001110-->101100

Critical path delay $t_{cd} = 5.625\text{ns}$